Problem 7.14: Determine the discharges in the parallel pipes and the pressure at A if the pressure at point B is 8 N/cm^2 and at C the pipe is open to the atmosphere in Fig. 7.14. For all pipes the Manning's roughness coefficient will be n=0.016.

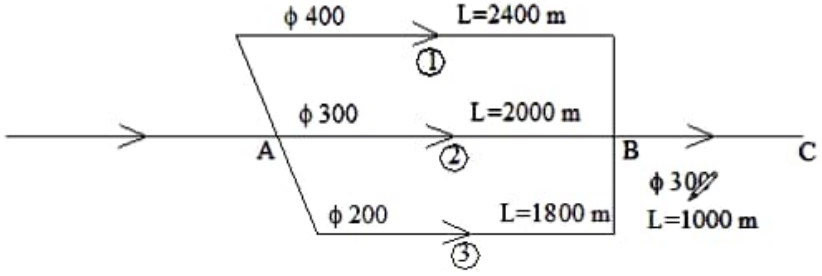


Figure 7.14

Solution:

$$\frac{P_B}{\gamma} = \frac{8\times10^{-3}/10^{-4}}{10}\frac{kN/m^2}{kN/m^2} = 8\ m$$

$$v = \frac{1}{n} \cdot R^{\frac{2}{3}} \cdot J^{\frac{1}{2}}$$

Energy equation between B-C:

$$\frac{P_B}{\gamma} = \frac{P_C}{\gamma} + \Delta h_{L_{BC}}$$

$$\frac{P_B}{\gamma} = 0 + \Delta h_{L_{BC}}$$

$$\Delta h_{L_{BC}} = 8\,\text{m}$$

The slope of the energy grade line $J_{BC}=\Delta h_{L_{BC}}=8/1000=0.008$

From Manning's equation

$$V_{BC} = \frac{1}{n} R_{BC}^{2/3} J_{BC}^{1/2} = \frac{1}{0.016}\left(\frac{0.3}{4}\right)^{2/3} (0.008)^{1/2} = 0.99\,\text{m/s} \approx 1\,\text{m/s}$$

$$Q_{BC} = V_{BC} A_{BC} = 1 \times \frac{\pi \times 0.3^2}{4} = 0.071\,\text{m}^3/\text{s}$$

Since the energy losses are equal in parallel pipes;

$$\Delta h_{L1} = \Delta h_{L2} = \Delta h_{L2}$$

$$J = \frac{V^2 n^2}{R^{4/3}}$$

$$\Delta h_L = J \times L = \frac{V^2 n^2}{R^{4/3}} L$$

$$\frac{V_1^2 n^2}{R_1^{4/3}} L_1 = \frac{V_2^2 n^2}{R_2^{4/3}} L_2 = \frac{V_3^2 n^2}{R_3^{4/3}} L_3$$

$$\frac{V_1^2 n^2}{R_1^{4/3}} L_1 = \frac{V_2^2 n^2}{R_2^{4/3}} L_2 = \frac{V_3^2 n^2}{R_3^{4/3}} L_3$$

$$\frac{V_1^2}{\left(\frac{0.4}{4}\right)^{1.33}} \times 2400 = \frac{V_2^2}{\left(\frac{0.3}{4}\right)^{1.33}} \times 2000 = \frac{V_3^2}{\left(\frac{0.2}{4}\right)^{1.33}} \times 1800$$

$$V_1^2 = 1.22V_2^2 = 1.89V_3^2$$

or

$$V_1 = 1.1V_2 = 1.37V_3 \qquad (1)$$

Continuity equation

$$Q = Q_1 + Q_2 + Q_3$$

$$0.071 = V_1 \frac{\pi \times 0.4^2}{4} + V_2 \frac{\pi \times 0.3^2}{4} + V_3 \frac{\pi \times 0.2^2}{4}$$

$$0.071 = 0.1256V_1 + 0.071V_2 + 0.0314V_3 \qquad (2)$$

If the equality (1) is substituted into the equation

$V_1 = 1.37V_3$

$V_2 = 1.245V_3$

$0.071 = 0.1256(1.37\ V_3) + 0.071(1.245\ V_3) + 0.0314\ V_3$

$0.071 = 0.172\ V_3 + 0.0884\ V_3 + 0.0314\ V_3$

$V_3 = 0.24\,m/s \qquad Q_3 = 0.24 \times \frac{\pi \times 0.2^2}{4} = 0.0075\ m^3/s$

$V_2 = 0.30\,m/s \qquad Q_2 = 0.021\ m^3/s$

$V_1 = 0.33\,m/s \qquad Q_1 = 0.0414\ m^3/s$

The pressure at point A

The pressure at point A

$$\frac{p_A}{\gamma} = \frac{p_B}{\gamma} + \Delta h_{LAB} = 8 + 1.294 = 9.294\,\text{mss}$$

Problem 7.15: In the system shown in Fig.7.10 water is pumped from A to reservoir B, find the power of the pump and draw the energy grade line of the system. The pipe length, discharge pumped from the pump, diameters of the pipes, and Kutter numbers were given in the figure

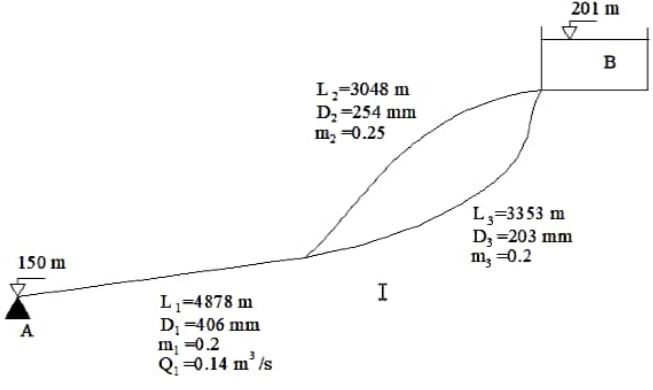


Figure p7.10a

Solution:

$$C = \frac{100\sqrt{R}}{m + \sqrt{R}}$$

$$C_1 = \frac{100\sqrt{0.406 / 4}}{0.2 + \sqrt{0.406 / 4}} = 61.43 \text{ m}^{1/2} / \text{s}$$

$$C_2 = \frac{100\sqrt{0.254 / 4}}{0.25 + \sqrt{0.254 / 4}} = 50.2 \text{ m}^{1/2} / \text{s}$$

$$C_3 = \frac{100\sqrt{0.203/4}}{0.2+\sqrt{0.203/4}} = 52.97\ m^{1/2}/s$$

$$V = C\sqrt{RJ} \quad \text{or} \quad J = \frac{V^2}{C^2R} \quad \text{and} \quad \Delta h_L = J \times L = \left(\frac{V^2}{C^2R}\right)L$$

In pipe (1)

$$Q_1 = 0.14 = V_1 \frac{\pi \times 0.406^2}{4}$$

$$V_1 = 1.08 m/s$$

$$\Delta h_{L1} = \frac{1.08^2 \times 4878}{61.43^2 \times \left(\frac{0.406}{4}\right)} = 14.85\,\text{m}$$

Since the head losses are equal in parallel pipe systems

$$\Delta h_{L2} = \Delta h_{L3}$$

$$\frac{V_2^2}{50.2^2 \times \left(\frac{0.254}{4}\right)} \times 3048 = \frac{V_3^2}{52.97^2 \times \left(\frac{0.203}{4}\right)} \times 3353$$

$V_3 = 0.89\ V_2$

Continuity equation

Continuity equation

$Q_1 = Q_2 + Q_3$

$$0.14 = V_2 \frac{\pi \times 0.254^2}{4} + (0.89V_2) \frac{\pi \times 0.203^2}{4}$$

$$V_2 = 1.76 \text{m/s}$$

$$V_3 = 1.57 \text{m/s}$$

$$\Delta h_{L2} = \frac{1.76^2 \times 3048}{50.2^2 \times \left(\frac{0.254}{4}\right)} = 59 \text{m}$$

The manometric height of the pump is obtained from the energy equations between A and B

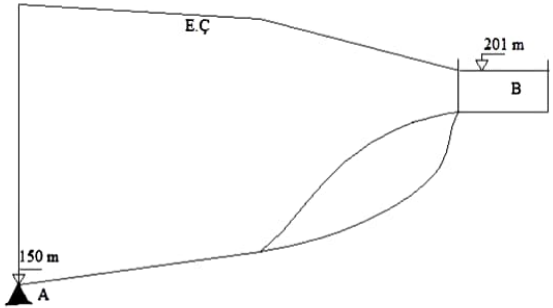
H_M=201-150+14.85+59=124.85 m

The power of the pump

$$N = \frac{\gamma(kN/m^3) \times Q(m^3/s) \times H_M(m)}{\eta} = \frac{10 \times 0.14 \times 124.85}{1} = 174.79 \quad kW$$

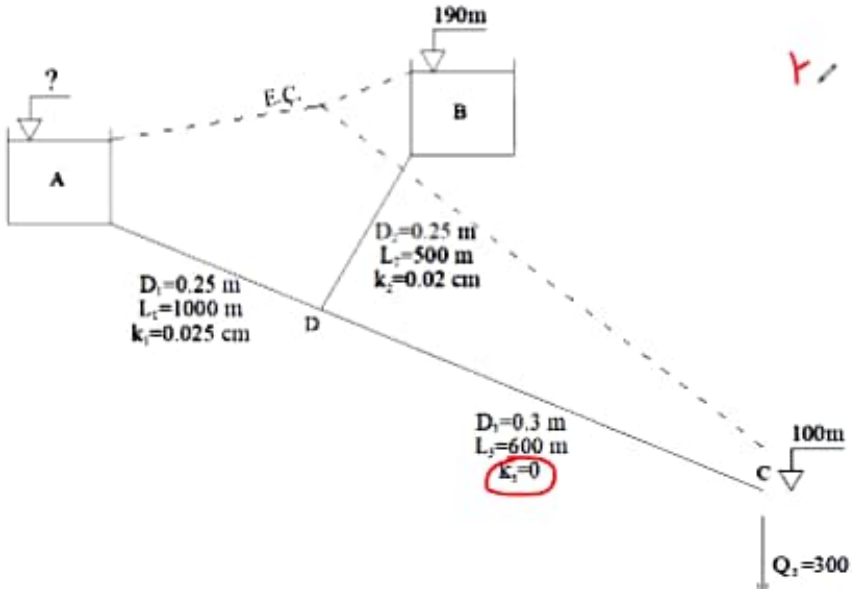
or

$$N = \frac{\gamma(kg/m^3) \times Q(m^3/s) \times H_M(m)}{75\eta} = \frac{1000 \times 0.14 \times 124.85}{75 \times 1} = 233.05 \quad BG$$


E.Ç
201 m
B
150 m
A

Problem 7.16: The discharge of the pipe DC is 300 lt/s and the pipe is open to atmosphere at C in the system shown in Fig. 7.16. If the outlet of the pipe is at 100m;

Find the elevation of the reservoir A and the discharges of pipes (1) and (2). Draw the energy grade line of the system.



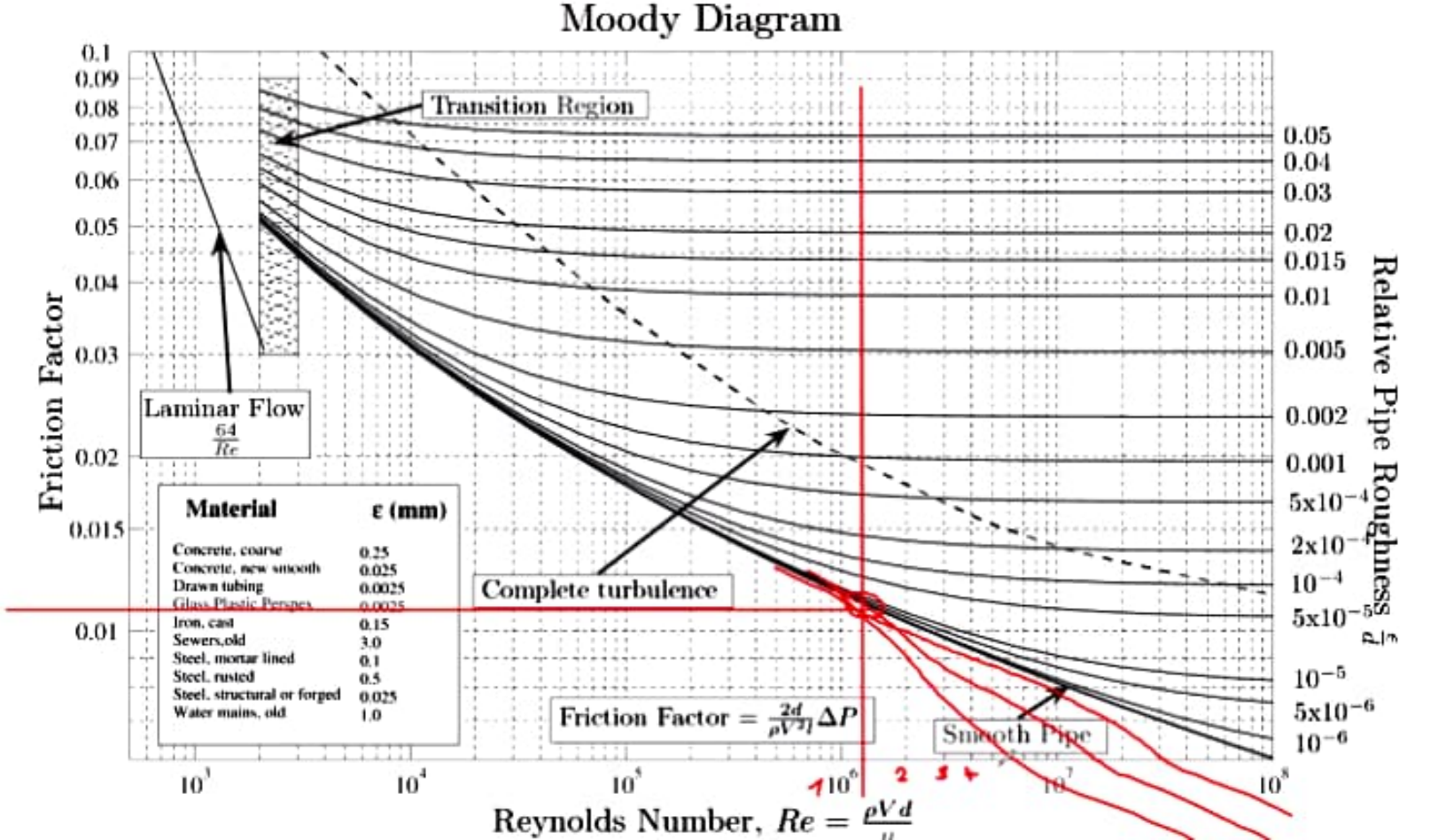
Solution:

$$0.3 = V_3 \frac{\pi \times 0.3^2}{4} \Rightarrow \qquad V_3 = 4.24 \text{ m/s}$$

$$\text{Re} = \frac{4.24 \times 0.3}{10^{-6}} = 1272000 \quad \text{and} \quad \frac{k}{D} = 0 \qquad \lambda = 0.011$$

$$\Delta h_{LDC} = 0.011 \frac{600}{0.3} \frac{4.24^2}{19.62} = 20.16$$

Moody Diagram



Material	ε (mm)
Concrete, coarse	0.25
Concrete, new smooth	0.025
Drawn tubing	0.0025
Glass, Plastic, Perspex	0.0025
Iron, cast	0.15
Sewers,old	3.0
Steel, mortar lined	0.1
Steel, rusted	0.5
Steel, structural or forged	0.025
Water mains, old	1.0

$$\left(z+\frac{P}{\gamma}\right)_D = 100+20.16 = 120.16 \quad ; \qquad \Delta h_{LBD} = 190-120.16 = 69.84\ m$$

$$\frac{k_2}{D_2} = \frac{0.02}{0.25} = 8\times10^{-4}$$

Assuming $\lambda = 0.019$

$$69.84 = 0.019 \frac{500}{0.25} \frac{V_2^2}{19.62} \Rightarrow \qquad V_2 = 6\ \mathrm{m/s}$$

Control;

$$\mathrm{Re} = \frac{6 \times 0.25}{10^{-6}} = 1.5 \times 10^{6} \quad \text{and} \quad \frac{k}{D} = 8 \times 10^{-4} \qquad \lambda = 0.0188 \cong 0.019 \ \textbf{(Suitable).}$$

$$Q_2 = 6 \times \frac{\pi \times 0.25^2}{4} = 0.294 \ \mathrm{m^3/s}$$

$$Q_1 = 0.0055 \ \mathrm{m^3/s}$$

$Q_1 \cong 0$ and $Q_2 \cong Q_3$ assuming there is no flow to reservoir A;

$$z_A = \left(z + \frac{P}{\gamma} \right)_D = 120.16 \ \mathrm{m}$$

Problem 7.17: Determine the discharge for each pipe in the network below by using Hardy-cross method (Consider two iterations). (n=2).

$$r_i = \frac{16}{2g\pi^2}\lambda_i \frac{L_i}{D_i^5}$$

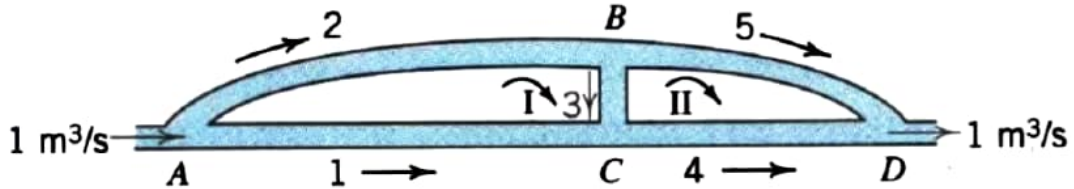


Figure 7.17

Pipe	Length (m)	Diameter (m)	Q_0 (m^3/s)	λ	r_i
1	1000	0.5	0.5	0.012	
2	1000	0.4	0.5	0.012	
3	100	0.4	0.4	0.012	
4	1000	0.5	0.9	0.012	
5	1000	0.3	0.1	0.013	

Solution:

Loop I

	rQ^2	nrQ
1	$-3*0.5^2$	$2*3*0.5$
2	$10*0.5^2$	$2*10*0.5$
3	$1*0.4^2$	$2*1*0.4$
∑	1.91	13.8

$$\Delta Q = -\left(\frac{1.91}{13.8}\right)$$

$$\Delta Q = -0.138\ m^3/s$$

Loop II

	rQ^2	nrQ
1	$-1*0.4^2$	2*1*0.4
2	$-3*0.9^2$	2*3*0.9
3	$46*0.1^2$	2*46*0.1
∑	-2.13	15.4

$$\Delta Q = -\left(-\frac{2.13}{15.4}\right)$$

$$\Delta Q = 0.138 m^3/s$$

DÜZELTİLMİŞ DEBİLER

1. Q_I = -0.5-0.138 = -0.638 m^3/s
2. Q_I = 0.5-0.138 = 0.362 m^3/s
3. Q_I = 0.4-0.138-0.138 = 0.124 m^3/s
3. Q_I = -0.4+0.138-(-0.138) = -0.124 m^3/s
4. Q_I = -0.9+0.138 = -0.762 m^3/s
5. Q_I = 0.1+0.138 = 0.238 m^3/s

II. İTERASYON

I.GÖZ

	rQ^2	nrQ
1	$-3*0.638^2$	2*3*0.638
2	$10*0.362^2$	2*10*0.362
3	$1*0.124^2$	2*1*0.124
∑	0.105	11.316

$$\Delta Q = -\left(\frac{0.105}{11.316}\right)$$

$$\Delta Q = -0.0093\ m^3/s$$

II.GÖZ

	rQ^2	nrQ
1	$-1*0.124^2$	2*1*0.124
2	$-3*0.762^2$	2*3*0.762
3	$46*0.238^2$	2*46*0.238
∑	0.848	26.716

$$\Delta Q = -\left(\frac{0.848}{26.716}\right)$$

$$\Delta Q = -0.032\ m^3/s$$

DÜZELTİLMİŞ DEBİLER

1. Q_{II} = -0.638-0.0093 = -0.647 m^3/s
2. Q_{II} = 0.362-0.0093 = 0.3527 m^3/s
3. Q_{II} = 0.124-0.0093-(-0.032) = 0.147 m^3/s
3. Q_{II} = -0.124- 0.032-(-0.0093) = -0.147 m^3/s
4. Q_{II} = -0.762-0.032 = -0.794 m^3/s
5. Q_{II} = 0.238-0.032 = 0.206 m^3/s